Joshua Williams

Alternative Energy and Power Generation Engineering | Penn State | Class of 2027
(570)-854-6557 | joshuaw6714@gmail.com | Nescopeck, PA | LinkedIn

EDUCATION

B.S. in Engineering, Alternative Energy and Power Generation Aug 2023 - May 2027
Interdisciplinary coursework spanning mechanical engineering and electrical engineering.
Minor: Engineering Leadership
Penn State University: Hazleton: GPA: 3.50 - Honors Program, Tri-Alpha Honor Society
Coursework: (Mechanical & Electrical Engineering): Power Electronics, Solar Energy System Design, Fuel Cells and Electrochemistry, Strength of Materials, Thermodynamics II, Renewable Energy Systems, Modeling (Ansys/MATLAB)

TECHNICAL SKILLS

Software and Tools: AutoCAD | Fusion 360 | SolidWorks | MATLAB (Simulation) | Ansys (FEA/Simulation) | Arduino | Python | Microsoft Excel | 3D Printing / Rapid Prototyping
Engineering: CAD Modeling, RCA, BOM, Systems Integration, Mechanical Design, Power Electronics, Energy Conversion, Process Improvement, Quality Control, SOP Writing, LOTO Compliance
Programming: C/C++ (Arduino, ESP32), Python (Raspberry Pi)

PROFESSIONAL EXPERIENCE

Engineering Intern, Insteel Wire Products: Hazleton, PA May 2025 - Present
- Executed a complete engineering cycle on production components: failure analysis, CAD redesign in Fusion 360, physical testing, reducing costs by up to 97% and generating over $10,000 in annual savings.
- Produced 100+ 2D technical drawings in AutoCAD and developed 75+ 3D CAD components
- Conducted FMEA on manufacturing processes, identifying failure modes and implementing corrective actions that reduced part defects and failures by over 50%.
- Authored a Quality Control Training Guide and revised SOPs and LOTO procedures (OSHA 1910.147) to standardize testing protocols, reduce operator error, and maintain safety compliance.

Manager, Broyan's Farm Market: Nescopeck, PA - December 2020 - August 2025
- Supervised and trained a team of 10+ employees over four years, managing daily operations and inventory control.

ENGINEERING PROJECTS

Autonomous Robotic Rover: Hardware Lead May 2026 - Present
- Designed a **large-scale autonomous 4WD robotic rover** in Fusion 360, including the chassis, suspension, steering, and custom 3D-printed components.
- Integrated a Raspberry Pi 4, ESP32, encoder-equipped motors, BTS7960 motor drivers, and a custom 3S6P battery system while leading hardware assembly and testing.
- Collaborated on autonomous navigation and developed project documentation, CAD, BOM, and hardware validation.

Single-Axis Solar Tracking System: Penn State Hazleton Aug 2025 - May 2026
- Designed, built, and tested a single-axis solar tracker that delivered a validated 17.5% improvement in energy yield over a fixed-angle baseline, confirmed through controlled electrical output measurements at multiple angles.
- Awarded 1st Place, Spring 2026 Penn State Honors Research Symposium.

Bike Generator, Project Lead, Science and Engineering Club: Penn State Hazleton Aug 2025 - May 2026
- Led a student team to design and build a human-powered electrical generator within a $400 budget, engineering the full mechanical-to-electrical conversion system for use at campus open house demonstrations.

HONORS, AWARDS & PUBLICATIONS

- 1st Place - Spring 2026 Penn State Honors Research Symposium | Single-Axis Solar Tracking System
- Co-author, **"WIP: Workshop Activities Focused on Hands-On Abilities for Engineering Students."** Presented at the ASEE Annual Conference

LEADERSHIP

President, Science and Engineering Club, Penn State Hazleton: Aug 2025 - Present
- Directed a series of technical projects including a PCB design workshop, Arduino motion sensor builds, and a drone build integrating custom PCBs and 3D-printed components. Organized the annual Cardboard Boat Race engineering challenge.